

Classwork (7th July)

1. The Kaliganj Community Health Centre recently launched a “10,000 Steps a Day” fitness campaign to encourage physical activity among residents. As part of the evaluation, the health centre collected the average daily step counts (in thousands of steps) of two groups of participants over one week:

- Group A: Participants who joined the program through the local school network (mostly younger adults).
- Group B: Participants who joined through the community clinic (mixed age groups).

The recorded step counts (in thousands of steps) are:

Group A:

6, 7, 8, 9, 9, 10, 11, 12, 13, 14, 35

Group B:

4, 5, 6, 7, 8, 9, 10, 11, 12, 18, 22

- (a) Construct a stem-and-leaf plot for the step counts (considering Group A and Group B as a combined data). Clearly indicate the stem and leaf units. (3 marks)
- (b) For Group A, calculate the first quartile (Q_1), median, and third quartile (Q_3). (3 marks)
- (c) Using the quartiles from part (b), determine the interquartile range (IQR) and identify any potential outliers using the $1.5 \times \text{IQR}$ rule. (2 marks)
- (d) Draw a boxplot for both groups on the same scale and briefly describe any differences in spread and possible outliers. (3 marks)
- (e) Calculate the range, standard deviation, and coefficient of variation (CV) for each group. Based on these measures, determine which group has greater relative variability in daily step counts. (4 marks)
- (f) Calculate Pearson's coefficient of skewness for Group B using

$$S_k = \frac{3(\bar{x} - \text{Median})}{s}$$

and comment on the direction of skewness. (3 marks)

- (g) Determine the 80th percentile (P_{80}) of Group A's step counts and explain what this value represents in the context of the fitness program. (2 marks)

2. The Bangladesh Digital Learning Initiative recently introduced an online learning platform for rural secondary school students. To evaluate access and engagement, the program recorded the average weekly internet usage (in hours) of 240 participating students.

The data are summarized below:

Weekly Internet Usage (hours)	Number of Students
0–5	18
5–10	36
10–15	54
15–20	62
20–25	40
25–30	22
30–35	8

- Construct a cumulative frequency table for the data. (2 marks)
- Calculate the median weekly internet usage. (3 marks)
- Calculate the first quartile (Q_1) and third quartile (Q_3). (4 marks)
- Using the quartiles, determine the interquartile range (IQR) and identify the lower and upper outlier boundaries. Comment on whether extreme internet usage values are likely in this dataset. (2 marks)
- Determine the 7th decile (D_7) and interpret its meaning in the context of students' online learning engagement. (2 marks)
- Calculate the variance and standard deviation of weekly internet usage. (3 marks)
- Compute the coefficient of variation (CV) and explain what it indicates about the relative variation in students' internet usage. (2 marks)
- Using Karl Pearson's coefficient of skewness

$$S_k = \frac{\bar{x} - \text{Mode}}{s}$$

determine whether the distribution is positively or negatively skewed. (2 marks)